

BabelDOC: Arbitrary Code Execution via CMap Pickle Deserialization in babeldoc/pdfminer/cmapdb.py

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1 Total Advisories	0 Critical	1 High	0 Medium
HIGH Severity	7.8 CVSS / Risk Score	TLP: CLEAR TLP Classification	PRO Customer Tier

Executive Summary

Arbitrary Code Execution via CMap Pickle Deserialization in babeldoc/pdfminer/cmapdb.py ### Summary BabelDOC's vendored PDF parser (`babeldoc/pdfminer/cmapdb.py`) deserializes untrusted pickle data when loading CMap files. The `_load_data()` method strips only NUL bytes from a PDF-controlled CMap name, then passes it directly to `os.path.join()` and `pickle.loads()`. Because Python's `os.path.join()` discards all preceding path components when it encounters an absolute path segment, an attacker who embeds a hex-encoded absolute path in a crafted PDF's `/Encoding` name (e.g., `2Ftmp2Fattacker2Fevil`) can redirect deserialization to any attacker-writable `.pickle.gz` file on the local system. Processing such a PDF results in arbitrary Python code execution with the privileges of the BabelDOC process. ### Details The vulnerable function is `CMapDB._load_data()` at `babeldoc/pdfminer/cmapdb.py:232-245`:

```
python @classmethod def _load_data(cls, name: str) -> Any: name = na
```

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Top Threat Advisories

High-priority advisories ranked by CVSS score, exploitability, and actor attribution.

Advisory Title	Severity	CVSS	EPSS	AI Summary
BabelDOC: Arbitrary Code Execution via CMap Pickle Deserialization in ba	Critical	7.0	-	## Arbitrary Code Execution via CMap Pickle Deserialization in ba ### Summary BabelDOC's vendored PDF p

MITRE ATT&CK® v15 Mapping

Technique density score: 0.0 · Techniques observed: 0 · Tactics covered: 0

MITRE ATT&CK techniques pending analyst enrichment. Deploy detection rules from Section 6 to identify related activity.

Threat Actor Intelligence

Attribution analysis across advisories in this report period.

Threat Actor	Count	Associated CVEs / Indicators
CDB-UNATTR-CVE	1	-

Indicators of Compromise (IOCs)

IOC extraction for this advisory is pending enrichment pipeline completion. PRO subscribers receive real-time IOC push to SIEM/SOAR via the SENTINEL APEX webhook at: GET /api/v1/intell/iocs?advisory={id}. Full IOC packages include IPv4, domain, SHA256, and URL indicators with confidence scores, first-seen timestamps, and STIX 2.1 formatted bundles.

Detection Engineering

Deploy these production-ready detection rules into your SIEM platform to identify exploitation attempts in real time. Compatible with Microsoft Sentinel, Splunk ES, and any Sigma-compatible platform.

Sigma Rule - Webserver / Process Monitoring

```
title: SENTINEL APEX - CDB-ADVISORY Exploitation Attempt
id: cdb-cdb-advisory-det
status: experimental
description: Detects exploitation indicators related to CDB-ADVISORY.
references:
- https://intel.cyberdudebivash.com
author: CyberDudeBivash SENTINEL APEX
date: 2026/07/11
tags:
- attack.initial_access
- attack.t1190
logsource:
category: webserver
detection:
keywords:
- 'CDB-ADVISORY'
condition: keywords
falsepositives:
- Security testing / penetration testing
level: medium
```

Microsoft Sentinel - KQL Query

```
// SENTINEL APEX - BabelDOC: Arbitrary Code Execution via CMap Pickle Deserialization in babeldoc/pdfminer/cmapdb.py
// Advisory | Microsoft Sentinel KQL
let time_window = ago(24h);
SecurityAlert
| where TimeGenerated > time_window
| where Description has "BabelDOC: Arbitrary Code Execution via C"
or AlertName has "BabelDOC: Arbitrary Code Execution via C"
or ExtendedProperties has "BabelDOC: Arbitrary Code Execution via C"
| extend Rule = "APEX-ADVISORY", Severity = "HIGH"
| project TimeGenerated, AlertName, AlertSeverity, Entities, Description, Rule, Severity
| order by TimeGenerated desc
```

Incident Response Playbook

Phase 1 - 0-24 Hours (Containment)

1. Activate IR team; assign lead analyst and executive sponsor.
2. Isolate or WAF-shield BabelDOC: Arbitrary Code Execution via CMap Pickle Deserialization in babeldoc/pdfminer/cmapdb.py instances exposed to the internet.

- 3. Enable verbose access logging on all BabelDOC: Arbitrary Code Execution via CMap Pickle Deserialization in babeldoc/pdfminer/cmapdb.py endpoints immediately.
- 4. Pull last 72h of web server and application logs for forensic baselining.
- 5. Check SENTINEL APEX IOC feeds and SIEM alerts for exploitation hits in your environment.
- 6. Notify relevant internal stakeholders and legal / compliance team.

Phase 2 - 24-72 Hours (Eradication)

- 1. Apply vendor patch for BabelDOC: Arbitrary Code Execution via CMap Pickle Deserialization in babeldoc/pdfminer/cmapdb.py or implement recommended mitigation.
- 2. Deploy Sigma and KQL detection rules from Section 6 across all SIEM platforms.
- 3. Rotate all API keys, service account credentials, and session tokens.
- 4. Conduct forensic timeline reconstruction for potential compromise window.
- 5. Validate patch deployment across 100% of affected instances via asset inventory.

Phase 3 - 7 Days (Recovery & Hardening)

- 1. Conduct post-incident review and update incident response runbooks.
- 2. Threat hunt for lateral movement or persistence artifacts using MITRE ATT&CK mapping.
- 3. Update asset inventory with patched version information and closure evidence.
- 4. Submit findings to internal risk register and CISO dashboard.
- 5. Schedule follow-up vulnerability scan to confirm remediation completeness.

Financial Impact Analysis (FAIR Model)

\$800K - \$4.5M Breach Cost Range (FAIR)	\$1.8M IBM Cost of Breach 2025 Median	\$450K/day Business Interruption Cost	\$320K Regulatory Fine Exposure
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High severity incidents activate IR retainers, forensic investigation (avg 47 days), and reputational damage impacting 12-18% of revenue.

Remediation Cost Estimate: \$180K · Includes engineering time, IR retainer activation, forensic investigation, and customer notification costs.

Regulatory & Compliance Implications

Organizations processing this advisory must evaluate obligations under applicable regulatory frameworks. The table below maps this threat to relevant compliance requirements.

Framework	Reference	Obligation
GDPR	Art. 32 / Art. 33	Appropriate technical measures; notify DPA if personal data affected

Framework	Reference	Obligation
PCI-DSS v4.0	Req. 6.3.3	Patches within 1 month; documented risk acceptance if deferred
NIS2 Directive	Art. 21	Risk management measures including vulnerability handling policies
NIST CSF 2.0	RS.MI / RC.RP	Incident mitigation and recovery planning; post-incident review

Actionable Recommendations

Prioritized defensive actions based on this period's intelligence.

1. Apply patches immediately for BabelDOC: Arbitrary Code Execution via CMap Pickle Deserialization in babeldoc/pdfminer/cmapdb.py.
2. Enable enhanced monitoring for related IOCs.
3. Review SIEM rules for associated MITRE ATT&CK techniques.
4. Apply vendor patch for BabelDOC: Arbitrary Code Execution via CMap Pickle Deserialization in babeldoc/pdfminer/cmapdb.py immediately - check NVD for official fix guidance.
5. Enable enhanced logging and alerting on all systems running affected software versions.
6. Deploy the Sigma detection rule from this report into your SIEM platform (Sentinel / Splunk / Elastic).
7. Audit all internet-facing instances of affected products and confirm patch deployment.

Appendix - Report Metadata & Legal

Field	Value
Report ID	intel--96e098b804966919efafebd8
Report Type	Weekly
Platform	CYBERDUDEBIVASH SENTINEL APEX v166.0 GOD-MODE
Generated At	2026-07-11T01:52:21.66266
Customer Tier	PRO
Customer Email	-
TLP	TLP: CLEAR
STIX Version	2.1
ATT&CK Version	v15

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